EDAB 6808| Assignment 2, Question 2

Forecasting South African Headline Inflation for 2026

A backtested comparison of ARIMA, damped-trend exponential smoothing and XGBoost

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Data: Statistics South Africa, CPI headline year-on-year inflation (P0141), January 2021 to December 2025

Summary

This report forecasts South Africa's headline consumer price inflation (year-on-year, %) for each month of 2026. Three models were trained on the 60 monthly observations from January 2021 to December 2025: an ARIMA(0,1,1) model, a damped-trend exponential smoothing (Holt) model, and a tuned XGBoost regressor with lagged and rolling features. All three were compared using an expanding-window, one-step-ahead backtest over 2025, with every tuning decision made on data before 2025.

XGBoost recorded the lowest MAE (0.204 percentage points) and MAPE (6.35%), Holt the lowest RMSE (0.266), and ARIMA the highest errors on all three measures. XGBoost was nonetheless kept as the primary model, and it projects inflation of roughly 3.5% throughout 2026 with a 95% interval widening from about 3.0–4.1% in January to 1.6–5.4% in December.

The first six months of 2026 have since been published. Inflation rose from 3.5% in January to 5.0% in June, and all three models missed the second-quarter acceleration. We report this as an out-of-sample check and did not use it for model selection.

1. Data collection and preparation

1.1 Source and forecasting target

The data is the official monthly headline CPI year-on-year inflation rates published by Statistics South Africa in its Consumer Price Index release (P0141). The training window is January 2021 to December 2025, which gives 60 monthly observations. The 2026 figures available at the time of writing (January to June) were read in but held out entirely from training and are used only in Section 7.3.

The target variable is the headline year-on-year inflation rate, not the CPI index level. The choice is used consistently in exploration, modelling, backtesting and the final forecast, and rests on four points:

- **Policy relevance.** The Reserve Bank targets an annual inflation rate rather than an index level, so a forecast of the rate can be compared directly with the target.
- **Interpretability.** A figure such as 3.5% is meaningful on its own, whereas an index value of 121.6 needs a base year and a previous value to interpret.
- **Consistency with the official series.** The rate is published directly, so no derived series with its own rounding is introduced.

- **Trend removal.** The index drifts upward with the price level, while the year-on-year rate is far more stable. This is also why the series shows little residual seasonality (Section 2).

1.2 Preparation and cleaning

The notebook visualised the years in rows and months in columns. It was reshaped to a single monthly series indexed by the first day of each month with a fixed month-start frequency, which statsmodels requires for date-aware forecasting. The window was then checked for completeness. There are no missing values, no duplicated timestamps and no implausible readings, so no imputation or outlier treatment was applied. Removing the 2022 peak, for example, would have discarded the most informative episode in the sample.

No differencing, logarithm or scaling was applied to the stored series. Differencing enters only inside the ARIMA model (d = 1), after the stationarity test in Section 2.3. Tree models are insensitive to feature scaling, so XGBoost was fitted on the raw percentages.

Table 1. Descriptive statistics for the training series (Jan 2021 to Dec 2025).

Observations	60	Mean	4.99%	Minimum	2.7% (Mar 2025)
Missing / duplicates	0 / 0	Std. deviation	1.51 pp	Maximum	7.8% (Jul 2022)
Skewness	0.14	Excess kurtosis	−1.01	Median	5.1%

2. Exploratory time-series analysis

2.1 Trend and level shifts

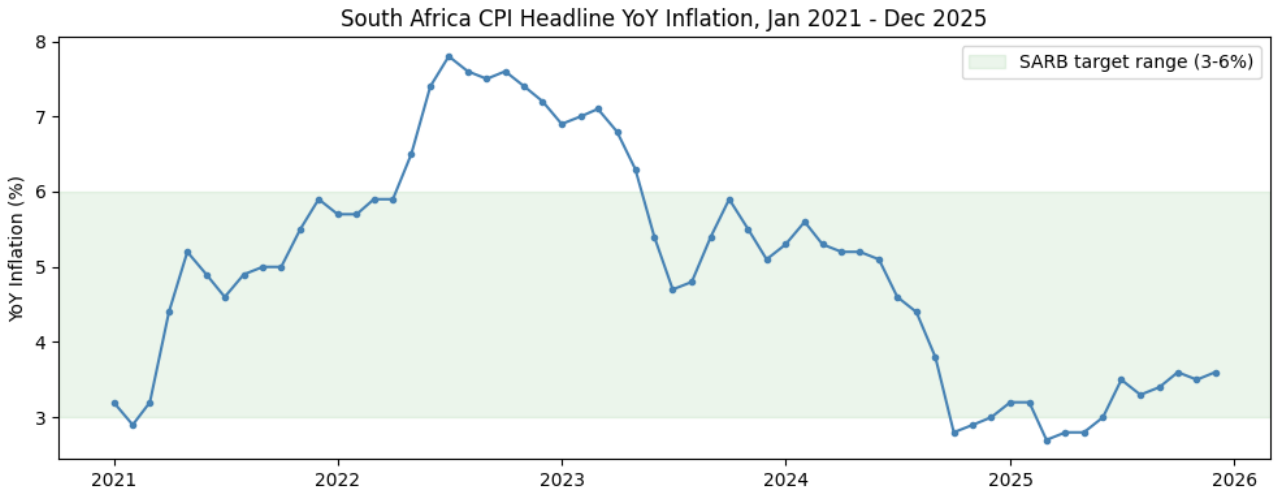


Figure 1. Headline CPI year-on-year inflation, with the former 3–6% target band shaded.

Inflation began 2021 near 3%, climbed through 2022 to a peak of 7.8% in July 2022, and stayed above the former 6% ceiling until mid-2023. It then fell steadily, reaching 2.8% in October 2024 and 2.7% in March 2025, and ended the sample at 3.6%. The series therefore contains one full up-and-down cycle, which is a small amount of history for any model to learn from.

The 3–6% band shown in Figure 1 was in force throughout the sample. In November 2025 the Minister of Finance and the Reserve Bank announced a new target of 3% with a tolerance band of one percentage point either side (2–4%) (Ministry of Finance and South African Reserve Bank, 2025). Since the forecast period falls under the new framework, the 2026 forecasts in Section 7 are judged against the 2–4% band.

2.2 Seasonality and volatility

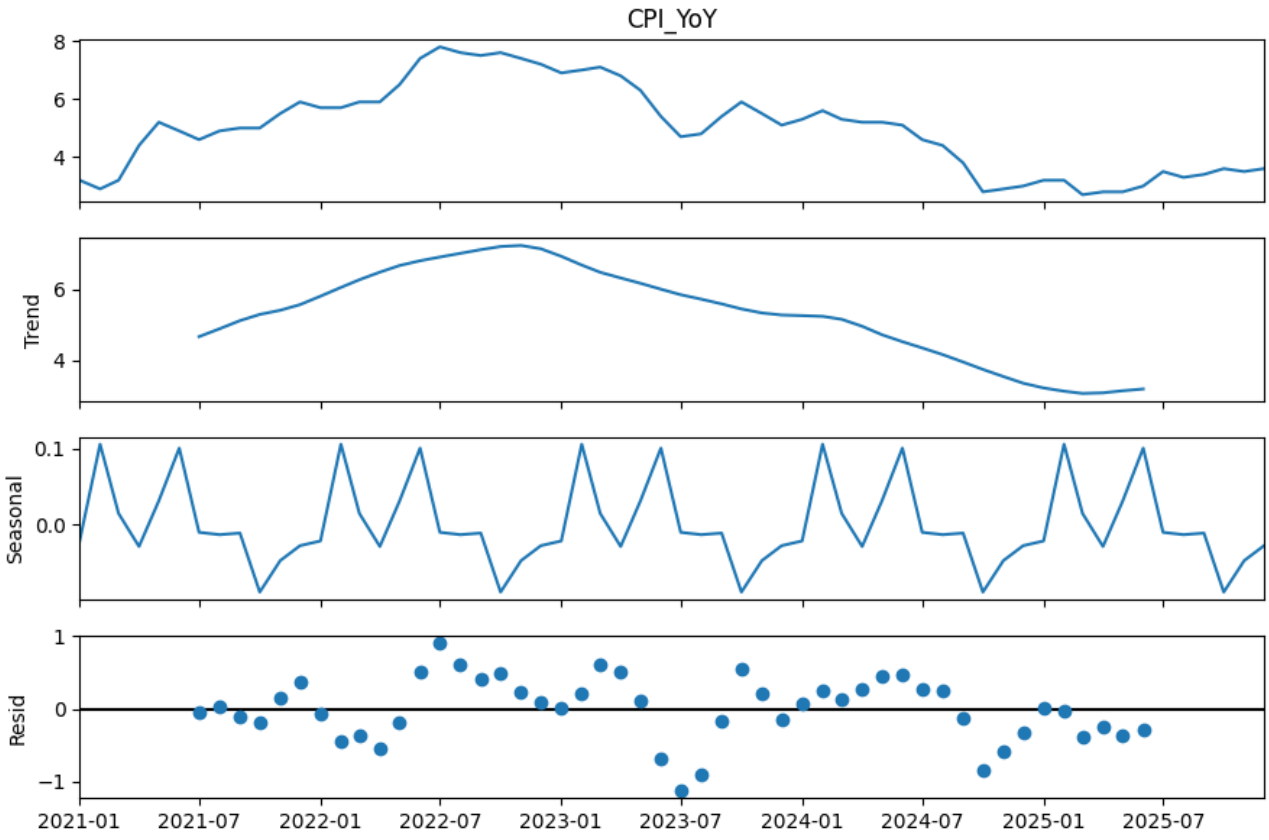


Figure 2. Additive decomposition (period 12) of the year-on-year series.

The decomposition assigns most of the variation to the trend component. The seasonal component is small relative to the trend, which is what one expects from a year-on-year rate: comparing each month with the same month a year earlier removes most of the regular calendar pattern. The residual is largest around the 2022 upswing and the 2023 to 2024 decline. Twelve-month rolling volatility (Appendix B) swings between roughly 0.3 and 1.05 pp as large moves enter and leave the window and is lowest in late 2025 when inflation sat between 3.0% and 3.6%. The series is therefore not homoscedastic, although with 60 points we do not attempt a volatility model.

2.3 Stationarity and autocorrelation

An Augmented Dickey–Fuller test on the level series gives a statistic of −1.210 ($p = 0.669$), so a unit root cannot be rejected. On the first difference the statistic is −5.809 ($p < 0.001$), and the differenced series is stationary. We therefore set $d = 1$. A second difference was rejected because over-differencing induces artificial negative autocorrelation, and a seasonal difference was rejected because the year-on-year transformation has already removed most seasonal structure. The ADF test has low power in short samples with

structural change, so d = 1 is supported by the residual diagnostics in Section 4 as well as by the test itself.

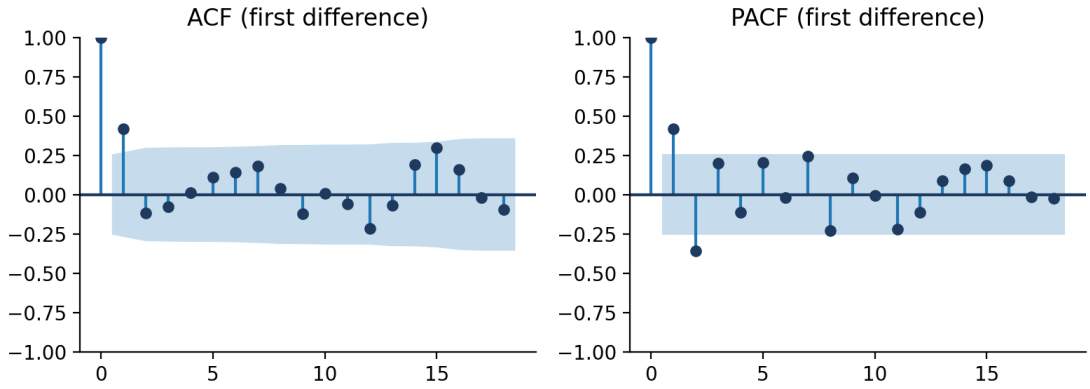


Figure 3. ACF and PACF of the first-differenced series.

The ACF and PACF of the differenced series show no clean cut-off, and there are no spikes at lag 12 in either plot, which is consistent with dropping seasonal terms. Since visual identification of (p, q) is ambiguous here, the orders were chosen by an information-criterion search instead (Section 4).

3. Backtesting design

A random train/test split would let the model see future observations when predicting the past, which is inappropriate for a persistent series. We use a time-aware, expanding-window (rolling-origin) design instead:

- **Evaluation window:** January to December 2025 (12 origins).
- **Procedure:** for each month in the window, refit the model on all data strictly before that month and forecast one step ahead, then move the origin forward by one month.
- **Tuning:** ARIMA orders and XGBoost hyperparameters were selected using only January 2021 to December 2024. XGBoost tuning used its own inner expanding-window validation on the last 12 months of that period, so the 2025 window was never used to choose a specification.
- **Leakage control:** all XGBoost features are built from lagged values (shift ≥ 1), so a prediction for month t uses only information available at t − 1.

All three models are evaluated under this identical design, so the metrics in Section 6 are directly comparable.

4. Statistical model: ARIMA

4.1 Why ARIMA

The series is non-stationary in levels but stationary after one difference, and shows no seasonal signal, so a non-seasonal ARIMA(p, 1, q) is the appropriate level of complexity. SARIMA was rejected because its seasonal terms are unsupported by the ACF and PACF and would consume scarce degrees of freedom. GARCH models conditional volatility of a return-like series, which is not the objective here.

4.2 Order selection and estimates

Every ARIMA(p, 1, q) with p and q from 0 to 3 was fitted to the pre-2025 data and ranked by AIC. ARIMA(0,1,1) was selected (AIC 35.17). Refitted to the full sample it gives the estimates below.

Table 2. ARIMA(0,1,1) estimates on the full 60-month sample.

MA(1) coefficient	0.816	Std. error 0.065	z = 12.48	p < 0.001
Innovation variance	0.104	Std. error 0.020	z = 5.13	p < 0.001
AIC / BIC	39.06 / 43.21	Log-likelihood	−17.53	60 observations

4.3 Diagnostics and interpretation

The residuals look like white noise. The Ljung–Box test does not reject independence (p = 0.69 at lag 1, 0.85 at lag 6 and 0.92 at lag 12), the Jarque–Bera test does not reject normality (p = 0.95, skewness −0.06, kurtosis 2.85), and the heteroskedasticity test is not significant (p = 0.87). The MA coefficient is large and highly significant.

An ARIMA(0,1,1) with θ = 0.816 is equivalent to simple exponential smoothing with a smoothing weight of about 0.18: the forecast is a slowly updated weighted average of past values, and it has no trend. Its multi-step forecast is therefore flat (3.69% for every month of 2026), and it can only widen its interval, not change direction. This is a property of the model class and should not be read as a claim that inflation will stay constant.

5. Machine-learning model: XGBoost

5.1 Features

The supervised dataset contains one row per month, built only from past values of the target. Twelve months are lost to the longest lag, so the usable sample runs from January 2022 (48 rows).

Feature	Role
Lags 1, 2, 3	Short-run persistence: inflation is strongly autocorrelated month to month.
Lag 6	Half-year memory, which picks up medium-term turning points.
Lag 12	Same month one year earlier, a check for any residual annual pattern.
3-month rolling mean	Smoothed recent level, less noisy than lag 1 alone.
3-month rolling std. dev.	Recent volatility, so the model can distinguish calm from turbulent spells.
Calendar month	Captures any residual seasonality left after the year-on-year transformation.

Table 3. XGBoost feature set. Rolling statistics are computed on the series shifted by one month.

5.2 Tuning and fit

A grid of 18 combinations (maximum depth 2, 3, 4; 100 or 200 trees; learning rate 0.03, 0.05, 0.1) was scored by expanding-window one-step MAE on the final 12 months of the pre-2025 data. The best setting was a maximum depth of 4, 100 trees and a learning rate of 0.1 (validation MAE 0.586), and it was used unchanged in the 2025 backtest. That validation MAE is much higher than the 2025 backtest MAE because the validation year, 2024, contained the sharpest fall in the series, which a tree model with a short history struggles to anticipate. The random seed was fixed at 42 for reproducibility.

5.3 Supplementary model: damped-trend exponential smoothing

Both ARIMA(0,1,1) and recursive XGBoost tend toward flat long-horizon forecasts. To add a trend-aware benchmark, we included Holt's additive-trend exponential smoothing with damping (Gardner & McKenzie, 1985), in which the trend contribution decays with the forecast horizon so it cannot extrapolate indefinitely. It was evaluated under the same expanding-window design.

6. Evaluation and model selection

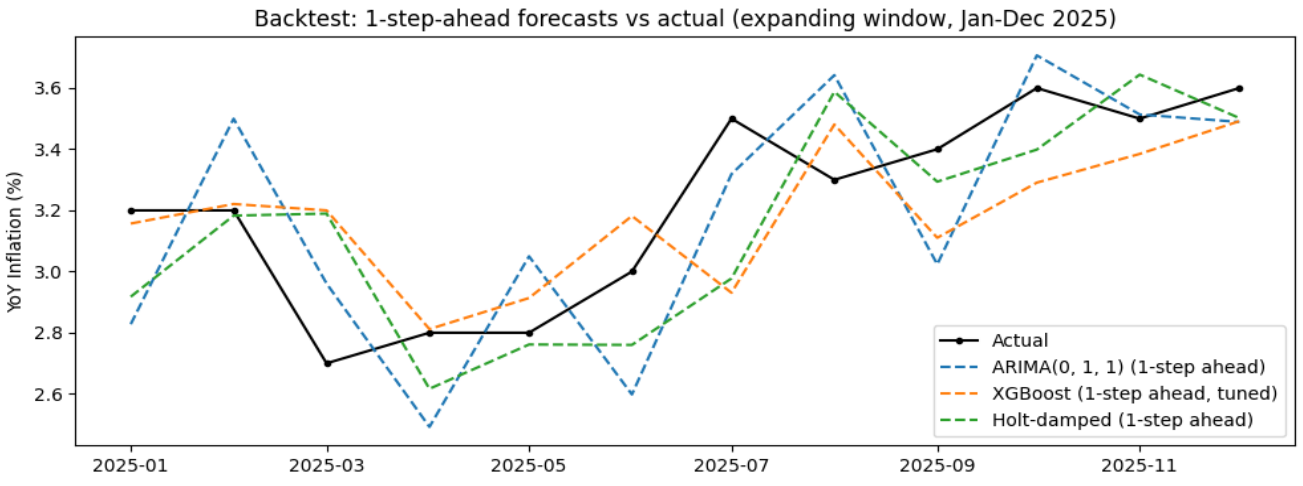
6.1 Metrics

MAE reports the average miss in percentage points and is the primary selection criterion. RMSE penalises large misses more heavily, which matters around turning points. MAPE is scale-free, which allows the models to be compared in relative terms. Classification measures such as precision and recall do not apply to a continuous target.

Table 4. Expanding-window one-step-ahead backtest, January to December 2025 (12 forecasts).

Model	MAE (pp)	RMSE (pp)	MAPE (%)	MASE
ARIMA(0,1,1)	0.252	0.278	8.09	0.740
XGBoost (depth 4, 100 trees, lr 0.1)	**0.204**	**0.268**	**6.35**	**0.598**
Holt damped trend	0.218	0.266	6.88	0.639

Figure 4. One-step-ahead backtest forecasts against actual inflation, 2025.



6.2 Which model performs best, and why

XGBoost has the lowest MAE and MAPE, and Holt has the lowest RMSE by a margin of 0.002 pp, which is negligible. ARIMA is weakest on every measure. Its one-step errors flip sign almost every month (for example −0.37 in January, +0.30 in February, −0.31 in April, +0.25 in May and −0.40 in June, forecast minus actual), which is the zigzag visible in Figure 4. This is what one expects from an MA(1) coefficient near 1: the forecast leans heavily on the previous forecast error, so a single surprise is partly reversed the next month. XGBoost is smoother and its errors are mostly small, but it misses at turning points. It overstated March's dip to 2.7% by 0.50 pp and missed the jump to 3.5% in July by 0.57 pp, and Holt shows the same two misses (+0.49 and −0.52). All three models were slow to react to those two months.

The evidence is weaker than the ranking suggests. The evaluation window has only 12 points, and MAE differs by 0.014 pp between XGBoost and Holt and 0.048 pp between XGBoost and ARIMA, differences that a single month's error could reverse. XGBoost is retained as the primary model because it satisfies the assignment's requirement to compare a statistical and a machine-learning method under one design and has the lowest MAE among them. We recommend treating the three forecasts as a range rather than trusting any one of them.

7. Final forecast for 2026

7.1 Refit and forecast

The three models were refitted on all 60 observations (January 2021 to December 2025) with the settings chosen in Sections 3 to 5 and used to forecast January to December 2026. ARIMA intervals come from the fitted model. XGBoost produces no analytical interval, so its 95% interval is the point forecast ± 1.96 times the standard deviation of the backtest residuals, scaled by the square root of the horizon. The same construction is applied to Holt. Multi-step XGBoost forecasts are recursive: each predicted value is fed back in as a lag for the next month.

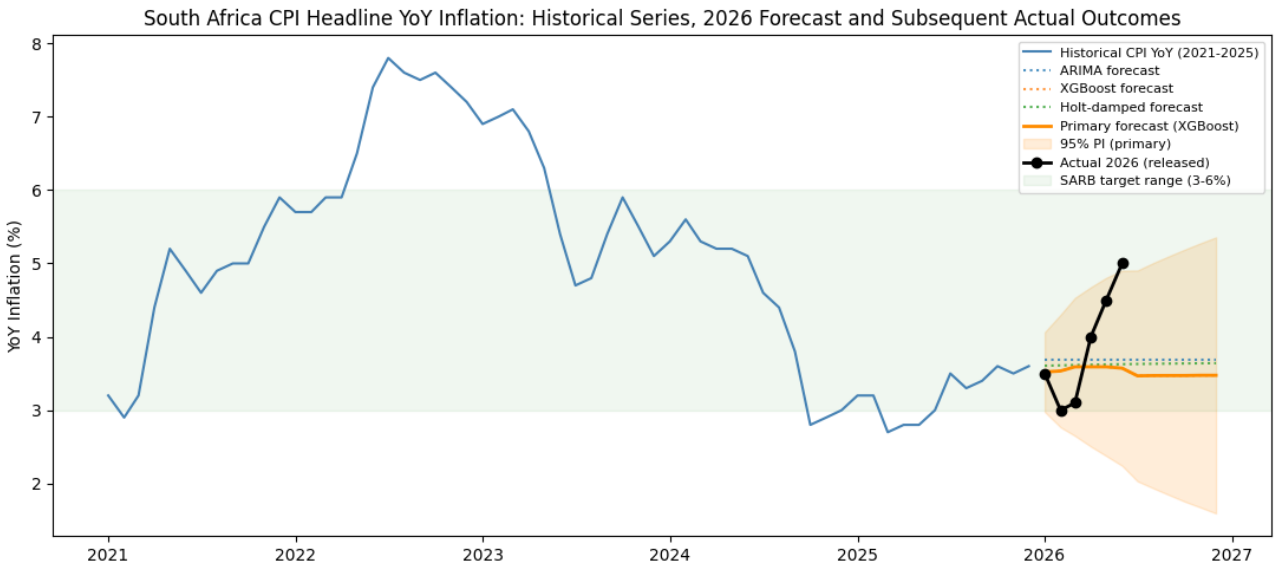


Figure 5. 2026 forecasts with the XGBoost 95% interval, and the January to June 2026 actuals (not used in fitting).

Month	XGBoost	Lower 95%	Upper 95%	ARIMA	Holt	Actual
Jan	3.52	2.98	4.06	3.69	3.61	3.5
Feb	3.53	2.77	4.30	3.69	3.61	3.0
Mar	3.59	2.65	4.53	3.69	3.62	3.1
Apr	3.59	2.50	4.68	3.69	3.62	4.0
May	3.59	2.38	4.81	3.69	3.62	4.5
Jun	3.57	2.24	4.90	3.69	3.63	5.0
Jul	3.47	2.03	4.90	3.69	3.63	4.3
Dec	3.47	1.59	5.36	3.69	3.64	n/a

Table 5. Selected 2026 forecasts (%). The full monthly table is in Appendix A and in the accompanying Excel file.

7.2 Interpretation

All three models place 2026 inflation between about 3.5% and 3.7% with almost no movement across the year. That sits inside the new 2–4% tolerance band and slightly above its 3% midpoint. The flatness reflects how the models work: the final observations (3.5% to 3.6% in the last quarter of 2025) dominate the lags, and recursive forecasting keeps feeding those levels back in. The point forecasts encode an assumption of stability, not a prediction of it.

7.3 Out-of-sample check against 2026 releases

Actual inflation for January to June 2026 is available from the same Stats SA series. It was not used in fitting, tuning or selection, and it is reported here only to gauge how the forecasts fared.

Month	Actual	XGBoost	ARIMA	Holt	XGB error
Jan	3.5	3.52	3.69	3.61	−0.02
Feb	3.0	3.53	3.69	3.61	−0.53
Mar	3.1	3.59	3.69	3.62	−0.49
Apr	4.0	3.59	3.69	3.62	+0.41
May	4.5	3.59	3.69	3.62	+0.91
Jun	5.0	3.57	3.69	3.63	+1.43
MAE		**0.632**	**0.650**	**0.645**	

Table 6. Forecast against actual, January to June 2026 (%). Error is actual minus the XGBoost forecast.

The forecasts were close in January and moderately high in February and March, when inflation dipped to 3.0%. From April the actuals rose quickly, and by June the XGBoost error was 1.43 pp. June's 5.0% also exceeded the upper end of the 95% interval (4.90%), so the interval was too narrow in that month. The three models had almost identical six-month MAEs (0.63 to 0.65 pp), which reinforces that they share the same blind spot: none could anticipate a shock that had not yet appeared in past inflation. Commentary on the 2026 rise attributes it to a fuel-price shock linked to the conflict in the Middle East (Ecofin Agency, 2026), but our models contain no fuel, exchange-rate or oil variables, so we cannot verify or quantify that channel here.

8. Uncertainty and limitations

- **Short sample.** Sixty monthly points contain a single inflation cycle. Rare or extreme episodes are barely represented, so the intervals likely understate tail risk, and the June 2026 breach is consistent with that.
- **Interval construction.** The XGBoost interval is a heuristic. It assumes one-step backtest errors scale with the square root of the horizon, based on only 12 residuals. It is symmetric, so the lower bound (1.59% in December) is mechanically low. It should be read as an approximate range, not a formal confidence interval.
- **One-step evaluation, twelve-step use.** Models were ranked on one-month-ahead accuracy, but the deliverable is a 12-month path. Multi-step accuracy could rank the models differently, and recursive forecasts compound their own errors. A direct multi-horizon backtest would be a natural extension.
- **Tree models do not extrapolate.** XGBoost cannot predict outside the range of targets it has seen, and with only 48 training rows its predictions collapse toward recent levels.
- **Univariate information set.** Only past inflation is used. Fuel prices, the exchange rate, food prices, administered prices, interest rates and inflation expectations are excluded, and these are the usual sources of turning points. Adding them would need aligned monthly series and forecasts of the regressors themselves.
- **Regime change.** The target framework changed in November 2025. A model trained on 2021 to 2025 does not know that policy is now aimed at a lower level.

- **Small evaluation window.** The 12-point backtest cannot separate models whose MAEs differ by hundredths of a percentage point. A longer history or multiple evaluation windows would give a firmer ranking.

Forecasts should be refitted as new releases arrive, since a forecast made from December 2025 data ages quickly.

9. Conclusion

A leakage-safe pipeline for forecasting South African headline inflation was built, compared a statistical model (ARIMA), a machine-learning model (XGBoost) and a damped-trend smoother under one expanding-window backtest, and produced monthly forecasts and intervals for 2026. XGBoost had the lowest MAE and MAPE in the 2025 backtest and projects inflation of about 3.5% for 2026. The margin over the alternatives is small, so the forecasts are presented as a central scenario rather than a firm prediction. The first half of 2026 shows why: inflation rose to 5.0% while the models expected about 3.6%. The main improvement would be adding leading indicators, particularly fuel, exchange-rate and food-price data, and evaluating at longer horizons.

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